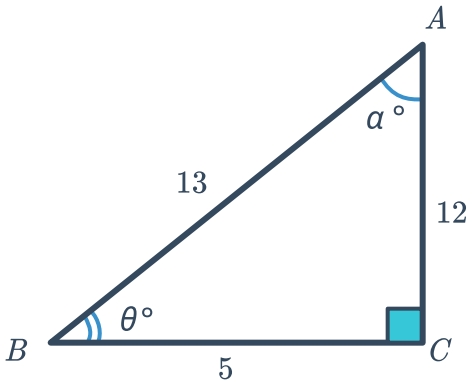
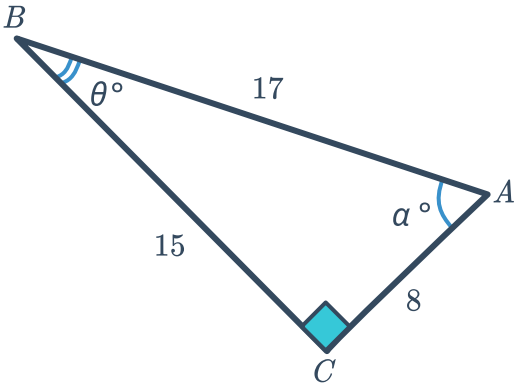
Trigonometric ratios

1 Evaluate $\cos\theta$ for each of the following triangles:

a

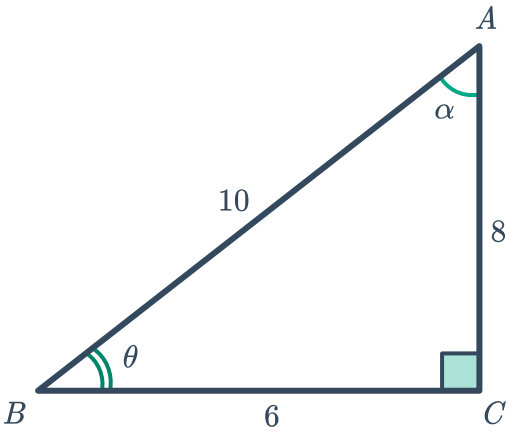


b

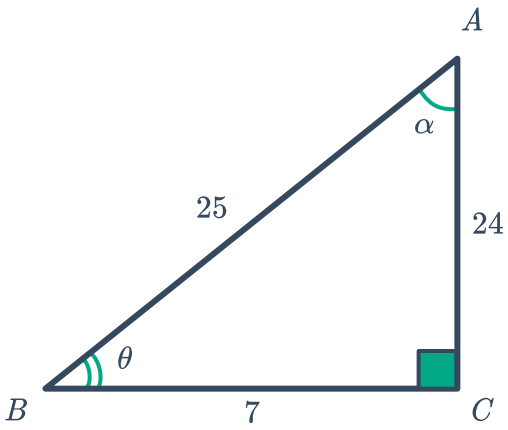


2 Evaluate $\sin\theta$ for each of the following triangles:

a

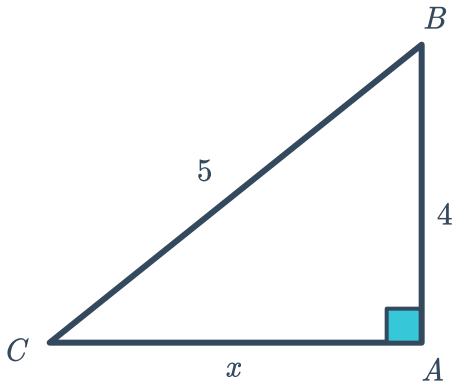


b



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3 In the following triangle, $\sin\theta = \frac{4}{5}$:

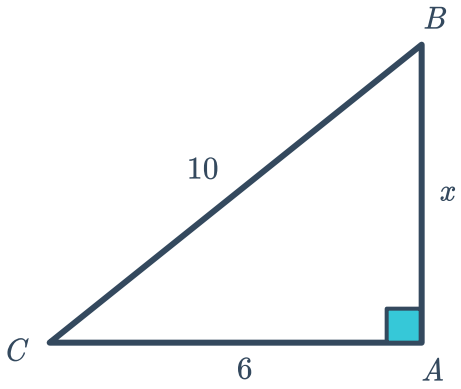


a Which angle is represented by θ?

b Find $\cos\theta$.

c Find $\tan\theta$.

4 In the following triangle, $\cos\theta = \frac{6}{10}$:

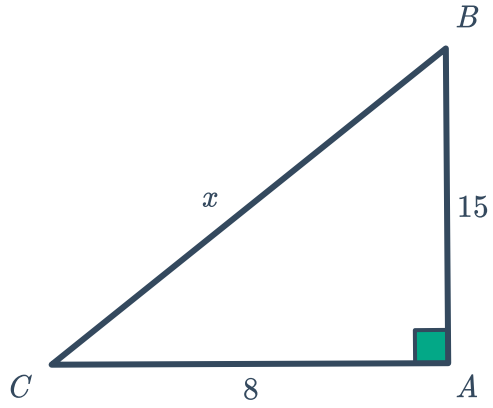


a Which angle is represented by θ?

b Find $\sin\theta$.

c Find $\tan\theta$.

5 In the following triangle, $\tan\theta = \frac{15}{8}$:



a Which angle is represented by θ?

b Find $\cos\theta$.

c Find $\sin\theta$.

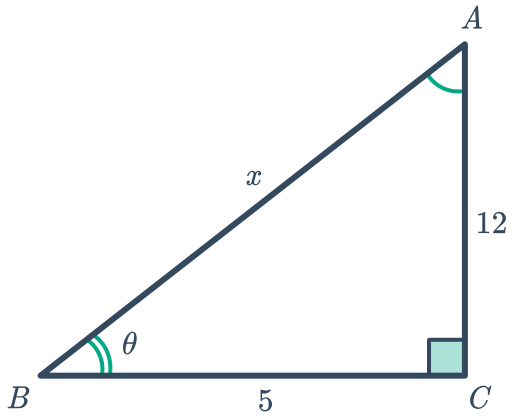
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6 Consider the following triangle:

a Find the value of x.

b Now, find $\sin\theta$.

c Now, find $\cos\theta$.

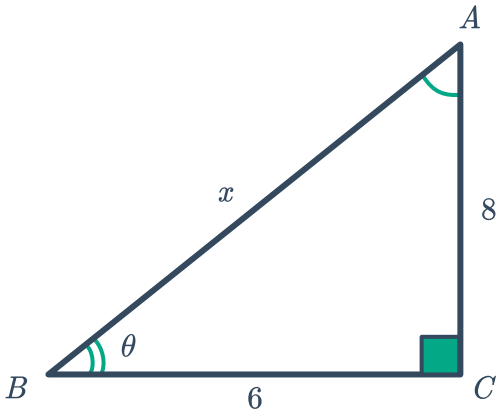


7 Consider the following triangle:

a Find the value of x.

b Find $\sin\theta$.

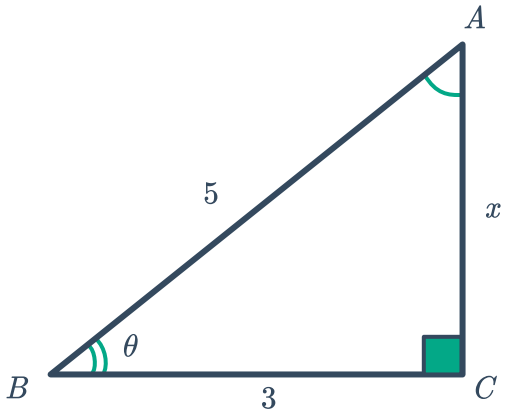
c Find $\cos\theta$.



8 Consider the following triangle:

a Find the value of x.

b Now, find $\tan\theta$.

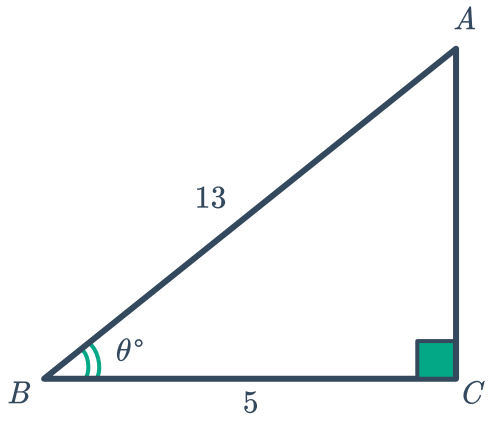


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9 Consider the following triangle:

a Find the length of AC.

b Find $\tan\theta$.

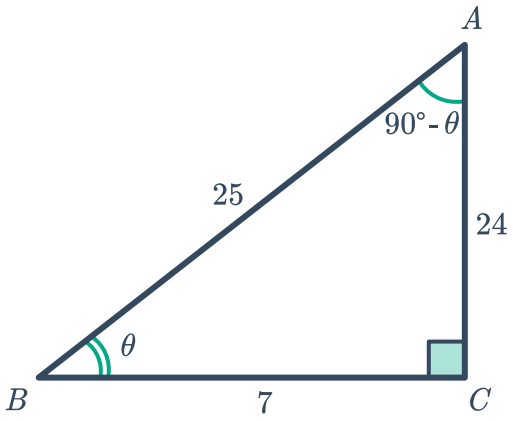


10 Consider the following triangle:

a Find $\cos(90^\circ - \theta)$ as a fraction.

b Find $\sin\theta$ as a fraction.

c What do you notice?

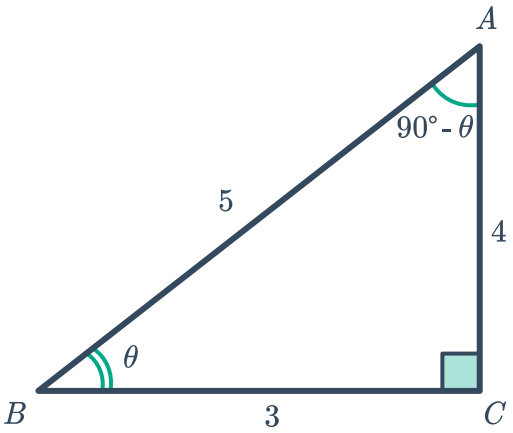


11 Consider the following triangle:

a Find $\sin(90^\circ - \theta)$ as a fraction.

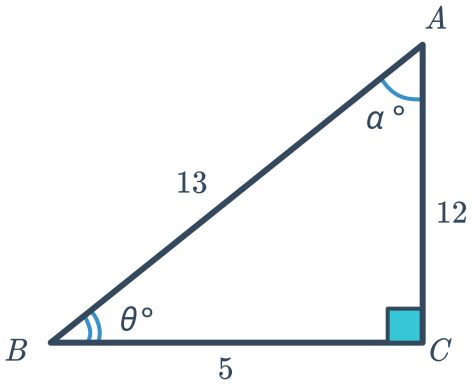
b Find $\cos\theta$ as a fraction.

c What do you notice?



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12 Consider the following triangle:



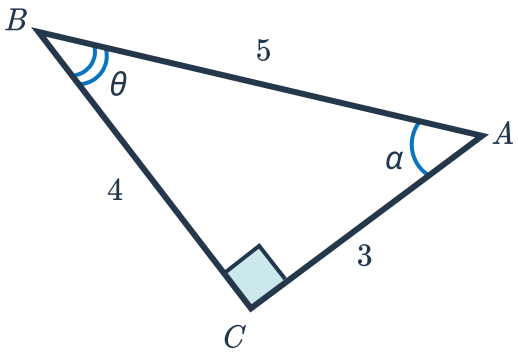
a Evaluate $\frac{\sin\theta}{\cos\theta}$.

b Evaluate $\tan\theta$.

c What do you notice?

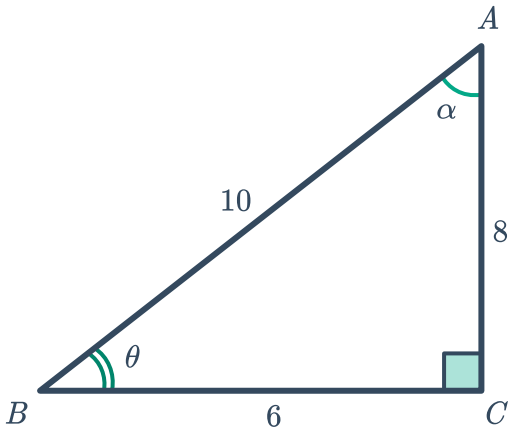
Reciprocal trigonometric ratios

13 Find $\sec\theta$ for the following triangle:

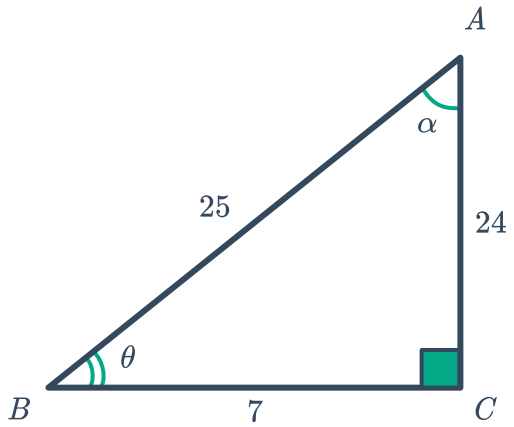


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14 Find $\csc\theta$ for the following triangle:



15 Find $\cot\theta$ for the following triangle:



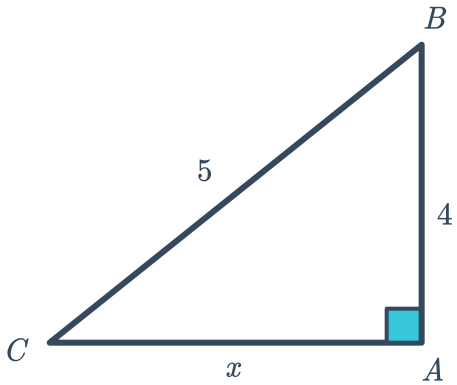
16 In the following triangle, $\csc\theta = \frac{5}{4}$:

a Which angle is represented by θ?

b Find the value of x.

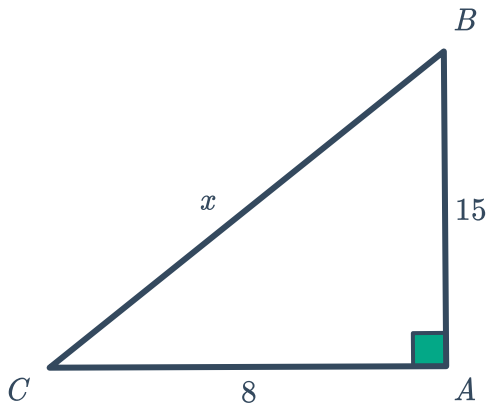
c Find $\sec\theta$.

d Find $\cot\theta$.



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17 In the following triangle, $\cot\theta = \frac{8}{15}$:



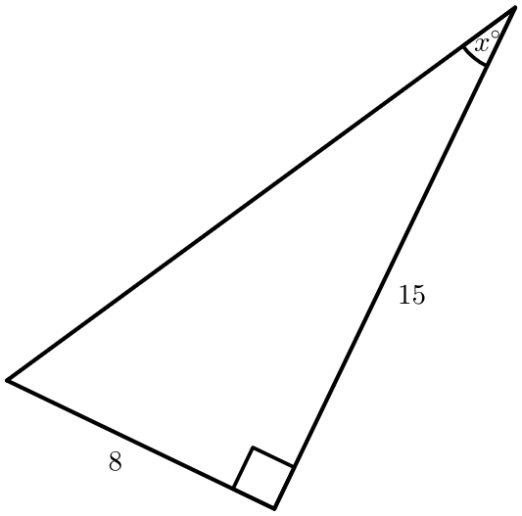
a Which angle is represented by θ?

b Find the value of x.

c Find $\sec\theta$.

d Find $\csc\theta$.

18 Consider the following triangle:



a Calculate the length of the missing side.

b Find the following trigonometric ratios:

i $\sin x$ **ii** $\cos x$

iii $\tan x$ **iv** $\sec x$

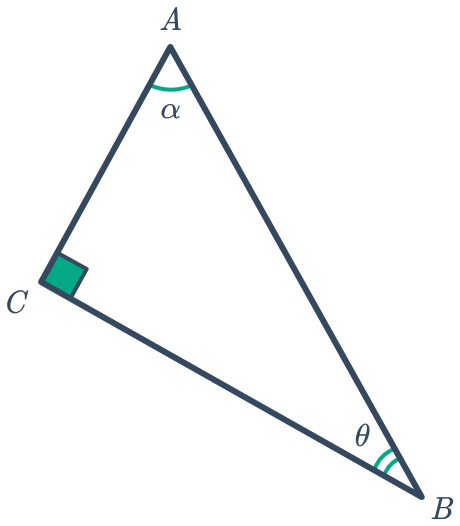
v $\csc x$ **vi** $\cot x$

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19 Find the indicated ratios for the following triangles:

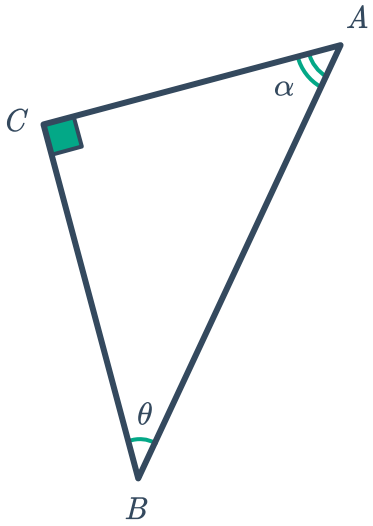
a

i $\tan\theta$ **ii** $\sin\alpha$



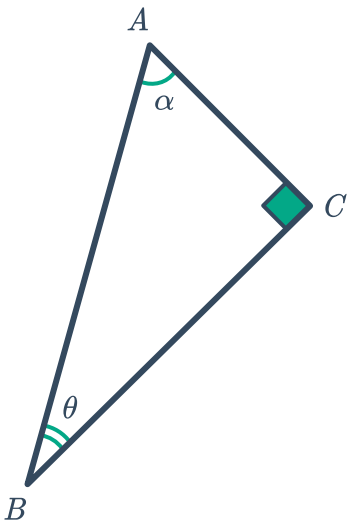
b

i $\sin\theta$ **ii** $\tan\alpha$



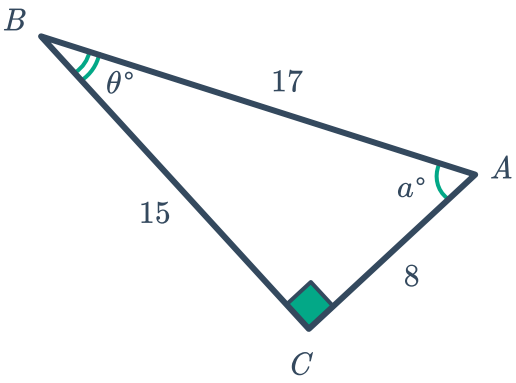
c

i $\cos\theta$ **ii** $\cos\alpha$



d

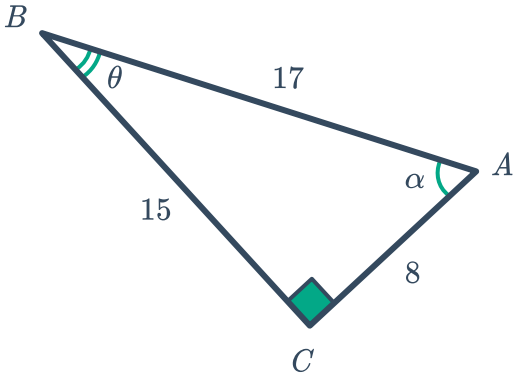
i $\tan\theta$ **ii** $\sin\alpha$



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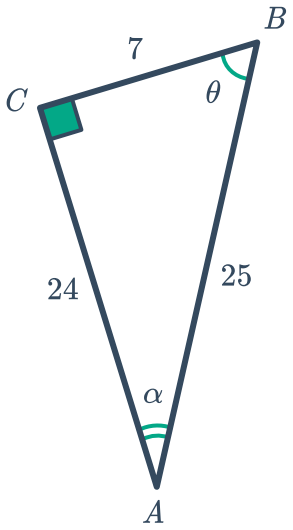
e

i $\cos\theta$ **ii** $\tan\alpha$

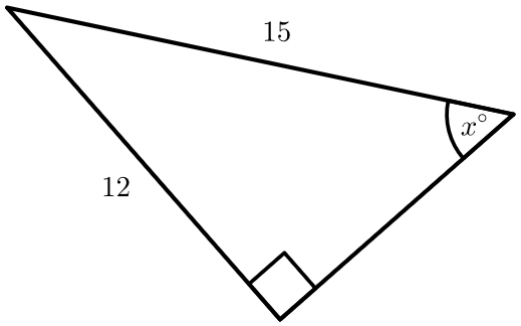


f

i $\sin\theta$ **ii** $\cos\alpha$



20 Consider the following triangle:



a Calculate the length of the missing side.

b Find the following trigonometric ratios:

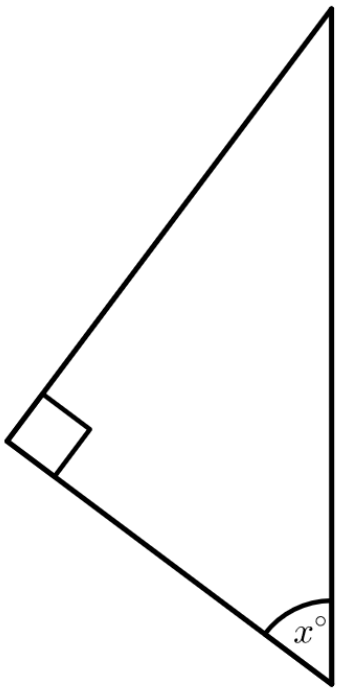
i $\sin x$ **ii** $\cos x$

iii $\tan x$ **iv** $\sec x$

v $\csc x$ **vi** $\cot x$

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21 Consider the following triangle:



a Given that $\tan x = \frac{3}{4}$, find $\cos x$.

b Given that $\sin x = \frac{15}{17}$, find $\tan x$.

c Given that $\cos x = \frac{12}{13}$, find $\sin x$.

d Given that $\sin x = \frac{4}{5}$, find $\csc x$.

e Given that $\tan x = \frac{7}{4}$, find $\cot x$.

f Given that $\cos x = \frac{5}{13}$, find $\sec x$.

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